Life Checkpoint: Circuits + Python

The only test is life. And this is life.

Name: _____ **Date:** _____

Class: _____ **Core mark:** / 44

No calculator is needed; the numbers are chosen to give clean answers. Work one line at a time. If one section stops you, continue to the next. If it all checks, you may pop off like you won the last stock in Smash Bros. Keep the controller on the desk.

A. Ohm's law and power (8 marks)

A 12 V test supply is connected across a 2.0 kΩ resistor. Show the formula, substitution, answer, and unit.

```
2.0 kΩ = __________ Ω

I = V / R = __________ / __________ = __________ A = __________ mA

P = V x I = __________ x __________ = __________ W = __________ mW
```

The resistor rating must be at least twice the calculated power. Circle one: 1/8 W 1/4 W

B. Draw a simple schematic (4 marks)

Draw one series loop containing a 9 V source, a closed switch S1, and resistor R1 = 1 kΩ. Use schematic symbols, label every part, show source polarity, and add one conventional-current arrow.

Your schematic

Page 1 mark: ____ / 12

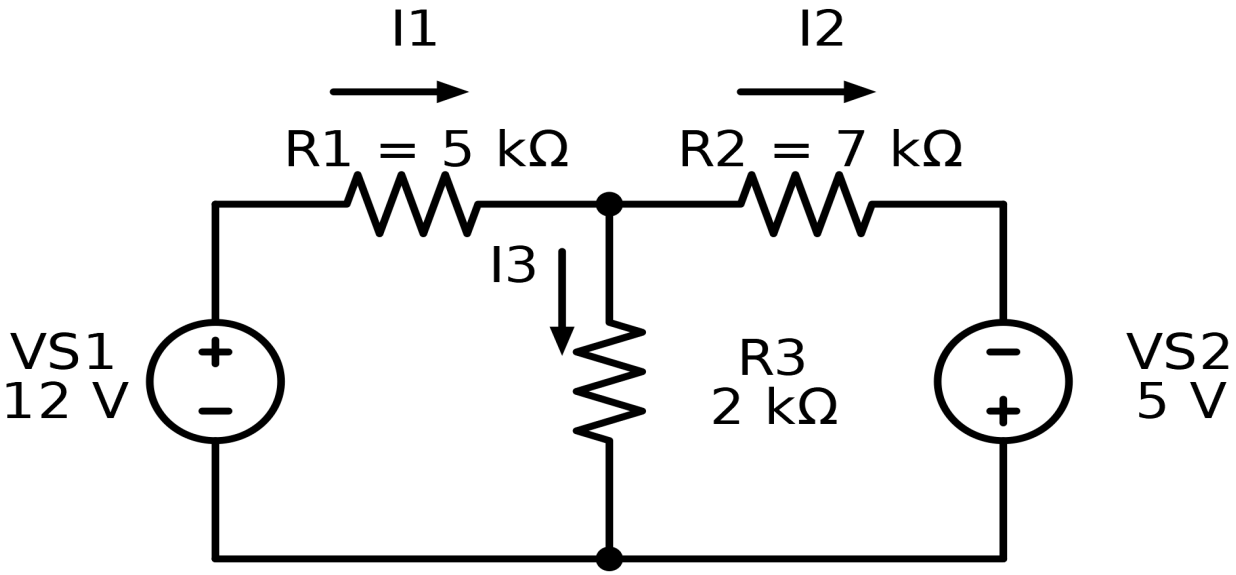
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C. Read and redraw the two-loop circuit (8 marks)

A controller test fixture uses two isolated supplies. Read the complete schematic, then redraw each loop by itself. Do not copy components that are not part of that loop.



1. Loop 1 only (3 marks)	2. Loop 2 only (3 marks)

3. Write KCL at the top centre node (2 marks)

I1 = _____ + _____

I3 = _____ - _____

Page 2 mark: ____ / 8

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D. KVL, standard form, and solution (12 marks)

1. Loop 1 (4 marks)

```
+12 - ______ I1 - ______ I3 = 0

+12 - 5I1 - 2( ___________________ ) = 0

Standard form: ______ I1 + ______ I2 = ______
```

2. Loop 2 (4 marks)

```
+5 - ______ I2 + ______ I3 = 0

+5 - 7I2 + 2( ___________________ ) = 0

Standard form: ______ I1 + ______ I2 = ______
```

RESTART POINT: If an earlier line stopped you, use 7I1 - 2I2 = 12 and -2I1 + 9I2 = 5 below.

3. Solve and check (4 marks)

```
I1 = ___________________ mA    I2 = ___________________ mA

I3 = I1 - I2 = ___________________ mA

One KCL or KVL check: ____________________________________________
```

E. Matrix bridge (4 separate marks)

Use variable order [I1, I2]. Copy one standard-form equation into each row.

Equation	I1 coefficient	I2 coefficient	Right side
Loop 1			
Loop 2			

```
[ ____  ____ ] [ I1 ]   [ ____ ]

[ ____  ____ ] [ I2 ] = [ ____ ]
```

Core page mark: ____ / 12 Matrix evidence: ____ / 4

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F. Python and NumPy (12 marks)

1. Complete the Ohm's-law program (4 marks)

Use the variable names already provided. Fill only the blanks.

```
voltage_V = 12

resistance_ohm = 2000

current_A = __________ / __________

power_W = __________ * __________

print(current_A)

print(power_W)
```

Write the two printed numbers: _____

2. Complete the NumPy array code (4 marks)

```
import numpy as np

voltages_V = np.__________([2, 4, 6, 8])

resistance_ohm = 2000

currents_mA = voltages_V / __________ * 1000

print(currents_mA)
```

Write the printed array: _____

3. Repair the type error (4 marks)

Circle the line that causes the error. Rewrite only that line, then write the output.

```
voltage_V = 12

resistance_ohm = "2000"

current_A = voltage_V / resistance_ohm

print(current_A)
```

Corrected line: _____

Printed output: _____

MARK RECORD: Circuits ___/32 Python ___/12 Core ___/44 Matrix ___/4

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